

Historical Note:

1. At the end of nineteenth century only classical physics, electromagnetic theory, and thermodynamics were known and scientific community were happy to explain the physical phenomena with the help of aforesaid theories.
2. At the beginning of the 20th century classical physics were seriously challenged on two major fronts: **(i) Relativistic domain:** where Newtonian mechanics ceases at very high speed (comparable to that of speed of light). **(ii) Microscopic domain:** when new experimental techniques evolved and scientists were able to dig out in microscopic domain (atomic and subatomic domains) of the materials, the classical physics were not able to explain experimental findings, like black body radiation, photoelectric effect, atomic stability, and atomic spectroscopy.
3. First attempt were made by Max Planck in 1900, when he was looking for the explanation of the experimental findings of black body. *In order to explain the black body radiation spectrum, he made a lucky guess and gave the postulate that the energy exchange between radiation and its surroundings takes place in discrete packets of energy called quanta.*

Quanta of Energy: *An electromagnetic wave of frequency ν Hz will have the corresponding energy $E = h\nu$ Joules called the quanta. Where $h = 6.626 \times 10^{-34}$ Joules.sec called the Planck constant.*

Planck gave accurate explanation of black body radiation and promoted new thoughts to see the new observations from different angles.

4. In 1905 Einstein explained the photo electric effect using the Planck's idea of energy quantization of electromagnetic wave and asserted that Planck idea of quantization of energy is valid for light as well. *He posited that light itself is made of discrete packets of energy particles called photon.*

5. In 1911 Rutherford did scattering experiment, in which he allowed an energetic alpha (α) particles to scatter with the gold foil. Based on the observation of this experiment he gave the atomic model:

Rutherford's atomic Model:

- (i) The mass of an atom is concentrated in a small space called the nucleus.
- (ii) Atoms majorly consist of positively charged particles.
- (iii) Negatively charged electrons revolve around nucleus (concentrated positive charge) in circular paths called orbits at very high speed.
- (iv) An atom is electrically neutral i.e. has no net charge.

Right after his atomic model, there was criticism regarding the stability of the atoms. As charged electrons revolve around nucleus in a circular path so it should experience acceleration due to which it should lose energy continuously in the form of electromagnetic radiations and then eventually fall into the nucleus thereby making the atom unstable. Also this model was not able to explain how the electrons are arranged in the orbits. *Although this model was not able to explain the atomic stability and electronic distribution in the orbits but it gave the basis of new thoughts which has become the basis of Quantum Mechanics.*

6. Niels Bohr Model:

- (i) Bohr's model suggested that electrons move around the nucleus of an atom in specific energy levels or orbits. These orbits are quantized, meaning that electrons can only occupy certain discrete energy levels and not any energy level in between.
- (ii) According to Bohr's theory, electrons can only exist in stable, stationary orbits with fixed energy values.
- (iii) Electrons can move between energy levels by absorbing or emitting specific amounts of energy in the form of photons (particles of light). When an electron absorbs energy, it moves to a higher energy level, and when it emits energy, it falls back to a lower energy level.

- (iv) Electrons in the innermost orbit have the lowest energy and are more stable, while those in the outer orbits have higher energy and are less stable. Electrons tend to occupy the lowest available energy levels, known as the ground state, but they can be excited to higher levels under certain conditions.
- (v) Bohr's model successfully explained the origin of spectral lines observed in the emission and absorption spectra of elements. When an electron moves between energy levels, it emits or absorbs photons of specific wavelengths, producing characteristic lines in the spectrum.

7. In 1923 Compton made an important discovery that made a most conclusive confirmation for the particle nature of electromagnetic wave/light.
8. This series of breakthrough given by Planck, Einstein, Bohr, and Compton gave both theoretical and experimental confirmation for the particle aspects of waves at the microscopic scale, where classical physics fails.
9. In 1923 Louis de Broglie postulated that electromagnetic radiation exhibits not only particle like behaviour but conversely material particles themselves display wave like behaviour. This concept was confirmed experimentally in 1927 by Davisson and Germer.
10. Davisson and Germer in 1927 showed that interference pattern a property of waves can be obtained with material particles such as electrons.
11. Owing all these experimental and theoretical facts necessity of the new theory was required and had prompted Heisenberg and Schrodinger to search for new theoretical foundation underlying these new ideas.
12. In 1925 Heisenberg gave his theory which was based on the Matrices and he expressed the quantities like energy, momentum, and angular momentum in terms of matrices and obtained an eigenvalue problem that describe the dynamics of microscopic systems.

- 13.** In 1926 Schrodinger's formulation of based on wave mechanics describes the dynamics of microscopic matter by means of wave equation called the Schrodinger equation. ***This is also called the fundamental principle of Quantum mechanics.***
- 14.** In 1927 Max Born proposed his probabilistic interpretation of wave mechanics.
- 15.** There were two approaches for the quantum mechanics (a) Heisenberg matrix approach (b) Schrodinger' wave formulation. Later Dirac suggested a general formalism of Quantum Mechanics which deals with abstract objects such as kets (state vectors) and bras and operators.

Wave particle duality

“Light is not only a wave but also a particle: Which means that light behaves like waves as well as particles.”

16.The Experiments which supports the particle nature of Light

- (i) *Blackbody radiation*
- (ii) *Photoelectric effect*
- (iii) *Compton effect*

17.The experiments which supports the wave nature of light.

- (i) *Young's double slit experiment*
- (ii) *Diffraction of light.*
- (iii) *Refraction of light*
- (iv) *Reflection of light*
- (v) *Polarization of light*

Particle Nature of light (Wave) – Blackbody radiation

Blackbody: A black body is an idealized body that absorbs all electromagnetic radiation incidents on it irrespective of its frequency or angle.

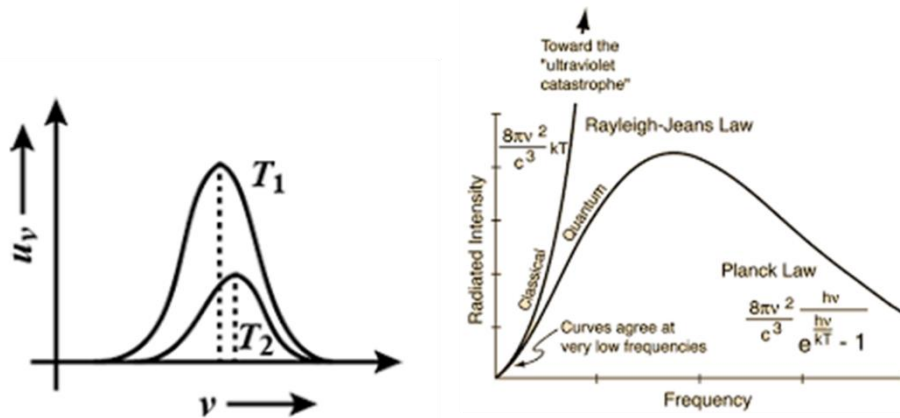


Fig. 1 Blackbody energy density spectrum.

If a black body is in thermal equilibrium at temperature T , it emits radiation at the same rate as it absorbs. The radiation emitted by the black body is known as blackbody radiation.

18. Rayleigh and Jeans explanation:

Suppose a blackbody cavity is in thermal equilibrium, the electromagnetic energy density inside the cavity can be obtained by multiplying the average energy of the oscillators (atomic oscillators present in the walls of the cavity) by the number of modes (standing waves in the cavity) of the radiation in the frequency interval ν and $(\nu + d\nu)$:

Number of modes per unit volume in the frequency range ν and $(\nu + d\nu)$:

$$N = \frac{8\pi\nu^2}{c^3} \dots \dots \dots (1)$$

Where c is the speed of light.

So the electromagnetic energy density in the frequency range ν and $(\nu + d\nu)$

$$u(\nu, T) = N\langle E \rangle = \frac{8\pi\nu^2}{c^3} \langle E \rangle \dots\dots\dots (2)$$

Where $\langle E \rangle$ is the average energy of the oscillators present on the cavity wall or the average energy of the radiations in the frequency range ν and $(\nu + d\nu)$

Average energy can be defined as

$$\langle E \rangle = \frac{\int_0^\infty E e^{-\frac{E}{k_B T}} dE}{\int_0^\infty e^{-\frac{E}{k_B T}} dE} = k_B T \dots\dots\dots (3)$$

So the equation (2) can be written as

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T \dots\dots\dots (4)$$

Where k_B = Boltzmann constant and T = Temperature of the black body.

Equation (4) is only true for lower frequencies. Further equation (4) predicts that for higher frequencies, energy density $u(\nu, T)$ attains higher values: which means for $\nu \rightarrow \infty$, the value of $u(\nu, T) \rightarrow \infty$, as it also shown in the figure (1). But in reality (experimentally observed spectrum) if $\nu \rightarrow \infty$, , the energy density $u(\nu, T) \rightarrow 0$. This discrepancy is known as **ultraviolet catastrophe.**

19. Max Planck explanation of Black body radiation:

In order to explain the black body radiation Max Planck in 1900 made a lucky guess and introduced a quanta of energy. He assumed that energy exchange between radiation and the surroundings takes place in terms of discrete packets of energy called quanta. He argued that electromagnetic radiation of frequency ν will exchange the energy to the surroundings (or matter) only in integral multiples of $h\nu$, ($E = nh\nu$ and $n = 1, 2, 3, \dots$) which is known energy packets or quanta.

Assuming the energy of the oscillator which emits electromagnetic radiation is quantized.

The average energy can be written as

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu e^{-\frac{nh\nu}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{nh\nu}{k_B T}}} = \frac{h\nu}{\left(e^{\frac{h\nu}{k_B T}} - 1\right)} \dots\dots\dots (5)$$

Having obtained the average energy of the oscillators he wrote the energy density of the black body:

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{\left(e^{\frac{h\nu}{k_B T}} - 1\right)} \dots\dots\dots (6)$$

Equation (6) is known as Planck's energy density formula for black body radiation.

Case I: for smaller frequencies $h\nu \ll k_B T$

$$e^{\frac{h\nu}{k_B T}} - 1 = 1 + \frac{h\nu}{k_B T} - 1 = \frac{h\nu}{k_B T} \dots\dots\dots (7)$$

After substituting the values of equation (7) in (6) we can get;

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T \dots\dots\dots (8)$$

Which is Rayleigh Jeans's law.

Case I: for higher frequencies $h\nu \gg k_B T$

$$\text{As } \nu \rightarrow \infty, \text{ the value } e^{\frac{h\nu}{k_B T}} \rightarrow \infty \text{ resulting } u(\nu, T) \rightarrow 0, \text{ from equation (6)}$$

Thus Planck has explained successfully black body radiation puzzle.

20. Energy density in terms of wavelength:

Let us rewrite the Planck energy distribution (Equation (6)) in terms of wave length λ .

Planck energy of the radiations in the frequency range ν and $(\nu + d\nu)$ from equation (6)

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{\left(e^{\frac{h\nu}{k_B T}} - 1\right)} \dots\dots\dots(6)$$

Since we know that

$$c = \nu\lambda \dots\dots\dots(9)$$

$$\Rightarrow \nu = \frac{c}{\lambda}$$

$$\Rightarrow \left| \frac{d\nu}{d\lambda} \right| = \left| \frac{c}{\lambda^2} \right|$$

Thus equation (6) can be written in terms of wavelength:

$$u(\lambda, T) = u(\nu, T) \left| \frac{d\nu}{d\lambda} \right| = \frac{8\pi c^2}{c^3 \lambda^2} \frac{hc/\lambda}{\left(e^{\frac{hc}{\lambda k_B T}} - 1\right)} \frac{c}{\lambda^2}$$

$$u(\lambda, T) = \frac{8\pi}{\lambda^5} \frac{hc}{\left(e^{\frac{hc}{\lambda k_B T}} - 1\right)} \dots\dots\dots(10)$$

21. Energy density in terms of angular frequency ω :

Since

$$\omega = 2\pi\nu \dots\dots\dots(11)$$

$$\left| \frac{d\nu}{d\omega} \right| = \frac{1}{2\pi} \dots\dots\dots(12)$$

$$u(\omega, T) = u(\nu, T) \left| \frac{d\nu}{d\omega} \right| = \frac{8\pi\omega^2}{(2\pi)^2 c^3} \frac{h\omega/2\pi}{\left(e^{\frac{h\omega}{2\pi k_B T}} - 1\right)} \frac{1}{2\pi}$$

$$u(\omega, T) = \frac{\hbar\omega^3}{(\pi)^2 c^3} \frac{1}{\left(e^{\frac{\hbar\omega}{k_B T}} - 1\right)} \dots\dots\dots (13)$$

22. Wein's displacement Law:

Wien's Displacement Law describes the wavelength at which the intensity of radiation emitted from a blackbody reaches its maximum point. After this point, the intensity decreases as temperature increases. This creates the characteristic shape of blackbody radiation curves.

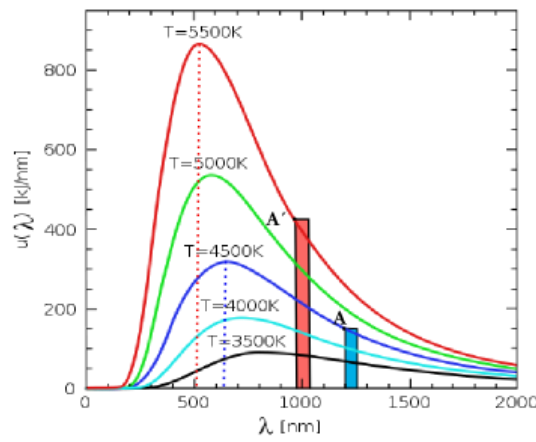


Fig. 2 Energy density verses wavelength curve.

As fig. 2 shows the emitted electromagnetic energy density at a particular temperature is maximum at a particular wavelength and after that the energy density decreases.

Figure shows that as the temperature is increased, the wavelength that is emitted with the greatest intensity becomes smaller. The wavelength will decrease linearly as the temperature is increased. Also the energy density peak shifted towards the lower wavelength side.

$$\lambda_{max} T = b \dots\dots\dots (14)$$

Where b is Wein's constant and $b = 2.8989 \times 10^{-3} \text{ meter} \cdot K$

Wein's law can be obtained by differentiating equation (10) with respect to λ and equating $\frac{du(\lambda, T)}{d\lambda} = 0$. Which will provide the corresponding wavelength at which the energy density emitted from the black body is maximum.

After differentiating equation (10) with respect to λ , we have

$$\frac{du(\lambda, T)}{d\lambda} = \frac{d}{d\lambda} \left(\frac{8\pi}{\lambda^5} \frac{hc}{\left(e^{\frac{hc}{\lambda k_B T}} - 1 \right)} \right) = 0$$

$$\Rightarrow 8\pi hc \left\{ \left(-5\lambda^{-6} \left(e^{\frac{hc}{\lambda k_B T}} - 1 \right)^{-1} \right) + \lambda^{-5} (-1) \left(e^{\frac{hc}{\lambda k_B T}} - 1 \right)^{-2} \left(\frac{hc}{k_B T} \right) \left(-\frac{1}{\lambda^2} \right) e^{\frac{hc}{\lambda k_B T}} \right\} = 0$$

$$\frac{8\pi hc}{\lambda^6} \times \left(e^{\frac{hc}{\lambda k_B T}} - 1 \right)^{-2} e^{\frac{hc}{\lambda k_B T}} \left\{ -5 \left(1 - e^{-\frac{hc}{\lambda k_B T}} \right) + \left(\frac{hc}{k_B T} \right) \frac{1}{\lambda} \right\} = 0$$

$$\text{Substituting } a = \left(\frac{hc}{k_B T} \right)$$

$$\left\{ -5 \left(1 - e^{-\frac{a}{\lambda}} \right) + \frac{a}{\lambda} \right\} = 0$$

$$\text{Taking } \frac{a}{\lambda} = 5 - \epsilon$$

We can obtained from above expression

$$\left\{ -5 \left(1 - e^{-(5-\epsilon)} \right) + (5 - \epsilon) \right\} = 0$$

After neglecting e^ϵ as it is very small.

$$\epsilon \approx 5e^{-5} = 0.0337$$

$$\frac{a}{\lambda} = 5 - 0.0337 = 4.9663$$

$$\frac{a}{\lambda} = \left(\frac{hc}{\lambda k_B T} \right) = 4.9663$$

$$\lambda_{max} T = b = 2.8989 \times 10^{-3} \text{ meter} \cdot K$$

Which is the Wein's Law expression.

23. Stefans-Boltzmann Law or Stefan's Law:

This law states that the total energy radiated from the black body per unit surface area per unit time is directly proportional to fourth power of black body's temperature T .

$$M = \sigma T^4 \dots\dots\dots(15)$$

Where $\sigma = 5.670 \times 10^{-8} W/m^2 K^4$ called Stefan-Boltzmann constant.

If the emitter is not blackbody then Stefan-Boltzmann law

$$M = \epsilon \sigma T^4 \dots\dots\dots(16)$$

Where ϵ is the emissivity of the emitting body and its value could be in the range

$0 \leq \epsilon \leq 1$. For blackbody radiator $\epsilon = 1$.

24. Photoelectric effect:

The first evidence for the particle nature of light was obtained in 1887 by Heinrich Hertz. He noticed in his experiment that upon the exposure of cathode by electromagnetic radiation charged particles were ejected from the cathode (target) surface. In 1900, by deflecting the emitted charges in a magnetic field, Lenard showed that the charges were electrons. This effect is called the photoelectric effect and the emitted photons are called photoelectrons.

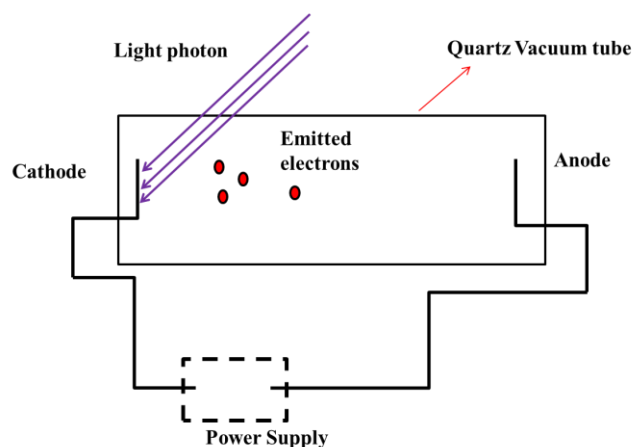


Fig. 3 Schematic of photoelectric effect setup.

Experimental findings of the photoelectric setup:

- a. The kinetic energy of the electrons is independent of the light intensity.
- b. For fixed light intensity, there is a maximum possible photocurrent, and maximum photocurrent is proportional to the light intensity falling on the cathode.
- c. The maximum kinetic energy of the photoelectrons depends on the frequency of the light but not on its intensity.
- d. There is a threshold frequency below which no photoelectrons are emitted. The threshold frequency is different for different metals.
- e. There is no time lag between incidence of the photon on the target and electron emission.

These results were not explained using classical physics.

Taking help of Max Planck quantization of energy, Einstein suggested that electromagnetic radiation is quantized, These quanta, which we now call photons, move without dividing and can only be emitted or absorbed as whole units. Einstein proposed that the energy of a photon is $E = h\nu$ which is related to the frequency of light. Where h is Planck's constant.

To explain the photoelectric effect, Einstein proposed that a single photon gives up its entire energy in liberating a single electron from the metal. In leaving the metal the electron gives up some of its energy to other electrons/atoms. The remaining energy appears as the kinetic energy of the electron. Conservation of energy requires

$$h\nu = \frac{1}{2}mv^2 + \phi \dots\dots\dots(17)$$

where ϕ is the energy needed to leave the metal and is called the **work function**.

$$\phi = h\nu_0 \dots\dots\dots(18)$$

Where ν_0 = threshold frequency: the minimum photon frequency required to release the electrons from the metal surface.